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Artificial Intelligence

Modules covered under the course

1. Introduction to Artificial Intelligence
2. Current and Future Industry Trends
3. Technical Roadmap to Artificial Intelligence

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Introduction to Artificial Intelligence

Understanding the Technology Beyond the Hype

NIELIT Ropar

Agenda

- AI – Introduction, what people think
- Define AI, what actually is AI
- Why AI matters today
- History, Turing test, Dartmouth Conference , ELIZA
- AI - The New Millennium
- Generative AI – chatgpt
- Scope of AI, risk & limitations
- Key take-aways, conclusion

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How people think about AI!



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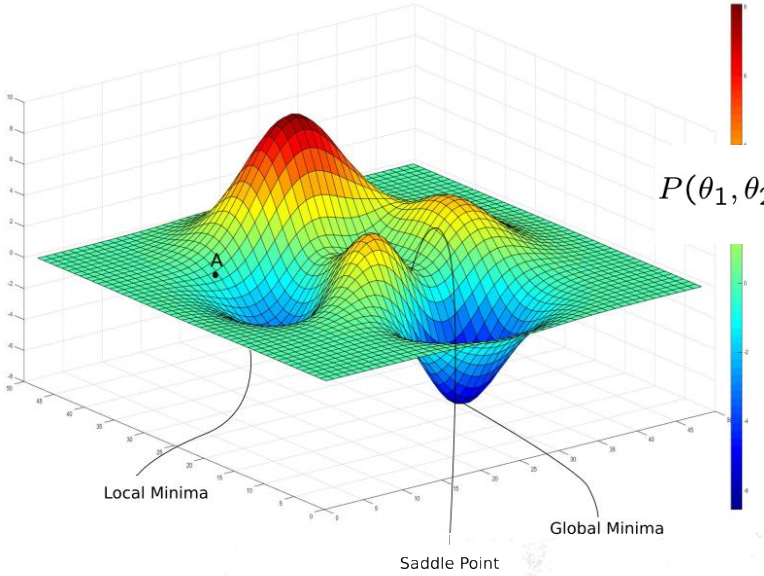
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What AI actually is!

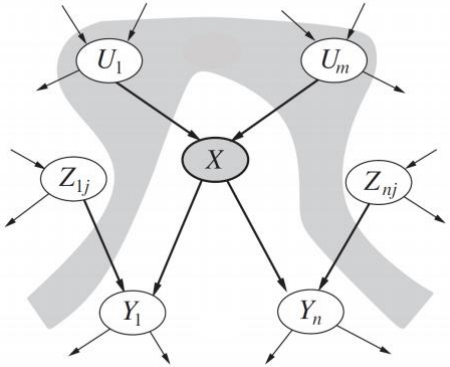
$$
\begin{aligned}
V^*(s) &= \max_a Q^{\pi^*}(s,a) \\
&= \max_a E_{\pi^*}\left\{ R_t \,\middle|\, s_t = s, a_t = a \right\} \\
&= \max_{a \in \mathcal{A}(s)} E_{\pi^*}\left\{ \sum_{k=0}^{\infty} \gamma^k r_{t+k+1} \,\middle|\, s_t = s, a_t = a \right\} \\
&= \max_{a \in \mathcal{A}(s)} E_{\pi^*}\left\{ r_{t+1} + \gamma \sum_{k=0}^{\infty} \gamma^k r_{t+k+2} \,\middle|\, s_t = s, a_t = a \right\} \\
&= \max_{a \in \mathcal{A}(s)} E\left\{ r_{t+1} + \gamma V^*(s_{t+1}) \,|\, s_t = s, a_t = a \right\} \qquad (3.14) \\
&= \max_{a \in \mathcal{A}(s)} \sum_{s'} \mathcal{P}^a_{ss'} \left[\mathcal{R}^a_{ss'} + \gamma V^*(s') \right], \qquad (3.15)
\end{aligned}
$$

$$
\frac{w}{\|w\|} \cdot (x_2 - x_1) = \text{width} = \frac{2}{\|w\|}
$$

$$
\begin{aligned}
w \cdot x_2 + b &= 1 \\
w \cdot x_1 + b &= -1 \\
w \cdot x_2 + b - w \cdot x_1 - b &= 1 - (-1) \\
w \cdot x_2 - w \cdot x_1 &= 2 \\
\frac{w}{\|w\|}(x_2 - x_1) &= \frac{2}{\|w\|}
\end{aligned}
$$



$$
P(\theta_1, \theta_2, \ldots, \theta_m) = \frac{\Gamma(\sum_k \alpha_k)}{\prod_k \Gamma(\alpha_k)} \prod_{k=1}^{m} \theta_k^{\alpha_k - 1}
$$



$$
\begin{aligned}
L(\mu, \sigma^2; x_1, \ldots, x_n) &= \prod_{j-1}^{n} f_X(x_j; \mu, \sigma^2) \\
&= \prod_{j-1}^{n} (2\pi\sigma^2)^{-1/2} \exp\left(-\frac{1}{2} \frac{(x_j - \mu)^2}{\sigma^2} \right) \\
&= (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{j-1}^{n} (x_j - \mu)^2 \right)
\end{aligned}
$$

Let us define A.I.



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A.I.



Expert Systems
Machine Learning
Symbolic AI
Robotics
Markov Models
Bayesian Networks
Classical Planning
Image processing
NLP
Pattern Recognition
CNN
Self-driving Cars
Autonomous Navigation
Computer Vision
Multi-agent Systems
LLMs
RNN
Decision Making
Problem Solving
Onotlogy
Speech Processing
Big Data
Deep Learning

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Why AI Matters Today

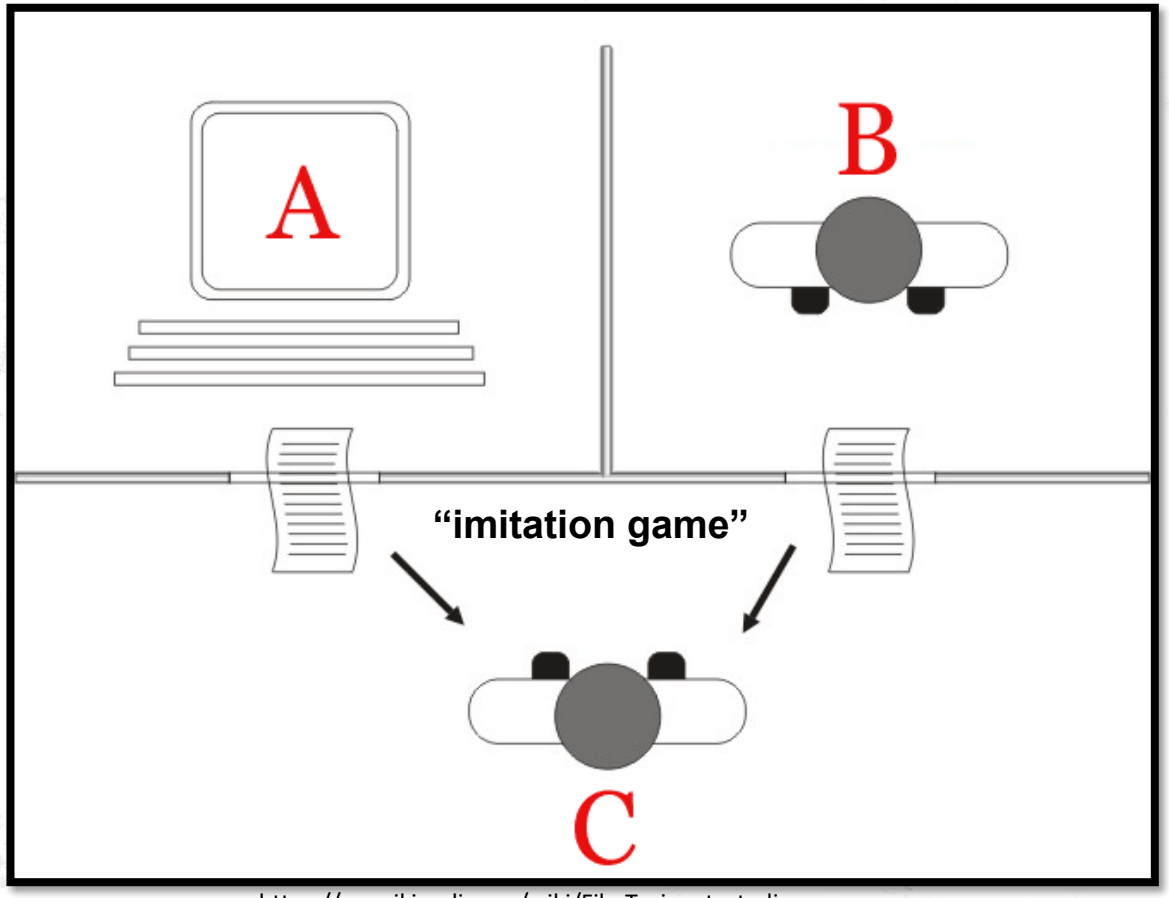
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- AI is transforming how work, services, and decisions are made:
  - It can handle scale beyond human capacity
  - It enables faster and more informed decisions
  - It improves consistency and efficiency
  - It can extract insights from complex data in limited time
- AI is becoming a **general-purpose technology**, like electricity or the internet.

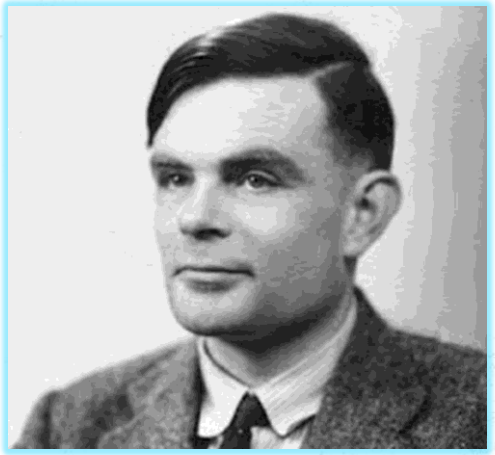
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The Turing test (1950)

- Developed by Alan Turing



https://en.wikipedia.org/wiki/File:Turing_test_diagram.png



The Loebner Prize

The Loebner Prize is an annual competition in artificial intelligence that awards prizes to the computer programs considered by the judges to be the most human-like. The format of the competition is that of a standard Turing test.

Dartmouth Conference (1956)

- Dartmouth Summer Research Project on Artificial Intelligence organized by John McCarthy



From left:
Trenchard More, John McCarthy, Marvin Minsky, Oliver Selfridge, and Ray Solomonoff

http://www.dartmouth.edu/~vox/0607/0724/ai50.html

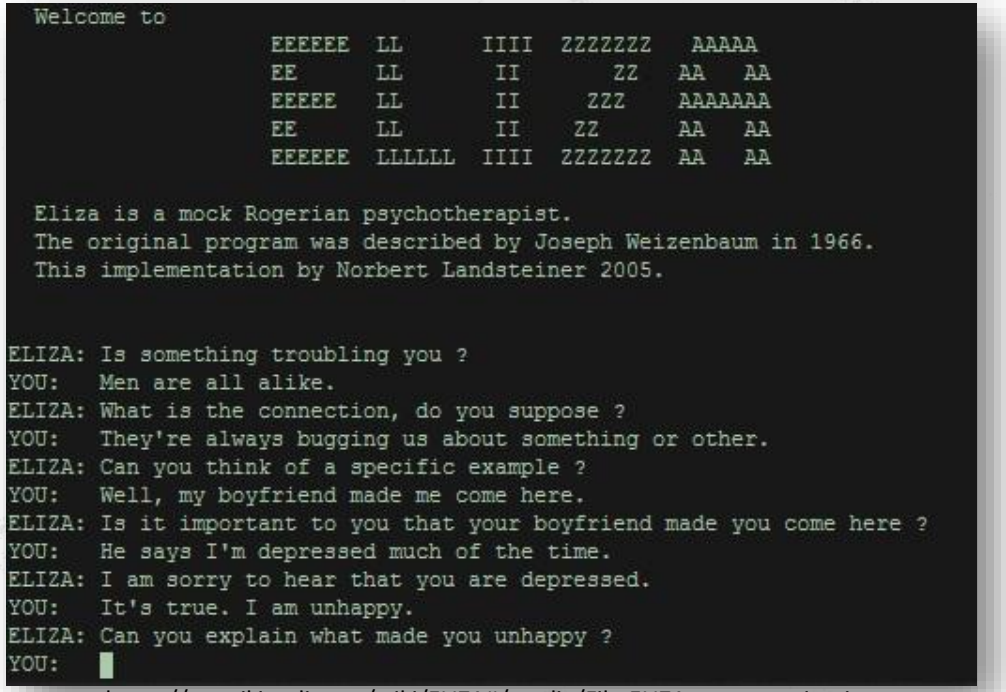
- Birth of the term "Artificial Intelligence (AI)"
- Gave AI, its mission, its first success and its major players

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The ELIZA Program (1964-66)

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- Created at the MIT Artificial Intelligence Laboratory by Joseph Weizenbaum
- "ELIZA - A Computer Program for the Study of Natural Language Communication between Man and Machine," Communications of the Association for Computing Machinery 9 (1966): 36-45.



https://en.wikipedia.org/wiki/ELIZA#/media/File:ELIZA_conversation.jpg



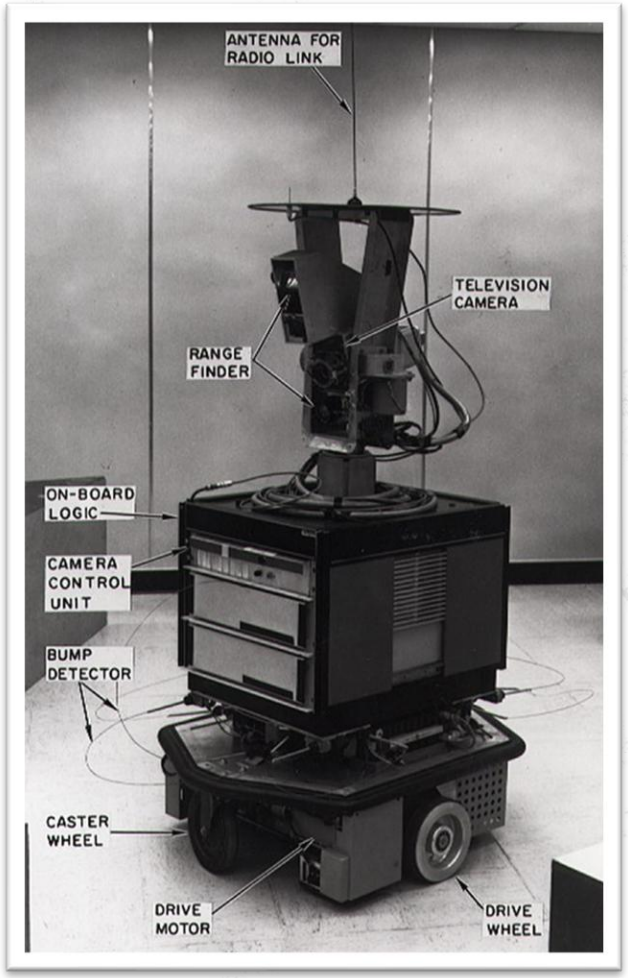
https://upload.wikimedia.org/wikipedia/commons/1/17/Joseph_Weizenbaum.jpg

Shakey the robot (1966-72)

- First general-purpose mobile robot to be able to reason about its own actions
- Combined research in robotics, computer vision, and natural language processing



https://en.wikipedia.org/wiki/Shakey_the_robot#/media/File:IEEE_Shakey_Team_at_dedication_of_IEEE_plaque_commemorating_their_achievement.jpg



https://en.wikipedia.org/wiki/Shakey_the_robot#/media/File:SRI_Shakey_with_callouts.jpg

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The First AI Winter(1974–1980)

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- Reasons of failures:
  - Over enthusiasm and very high claims as what AI can do
  - Limited Computing capability at that time
  - Only “toy” AI programs could be implemented
  - Limited availability of knowledge base
  - Shortage of funding



The Short Revival (1980–1987)

- Rise of Expert System in the industries (knowledge based systems)
- Japanese fifth generation computer project
- Revival of Connectionism
  - Discovery of back-propagation in Neural Networks
  - **Hopfield Networks**

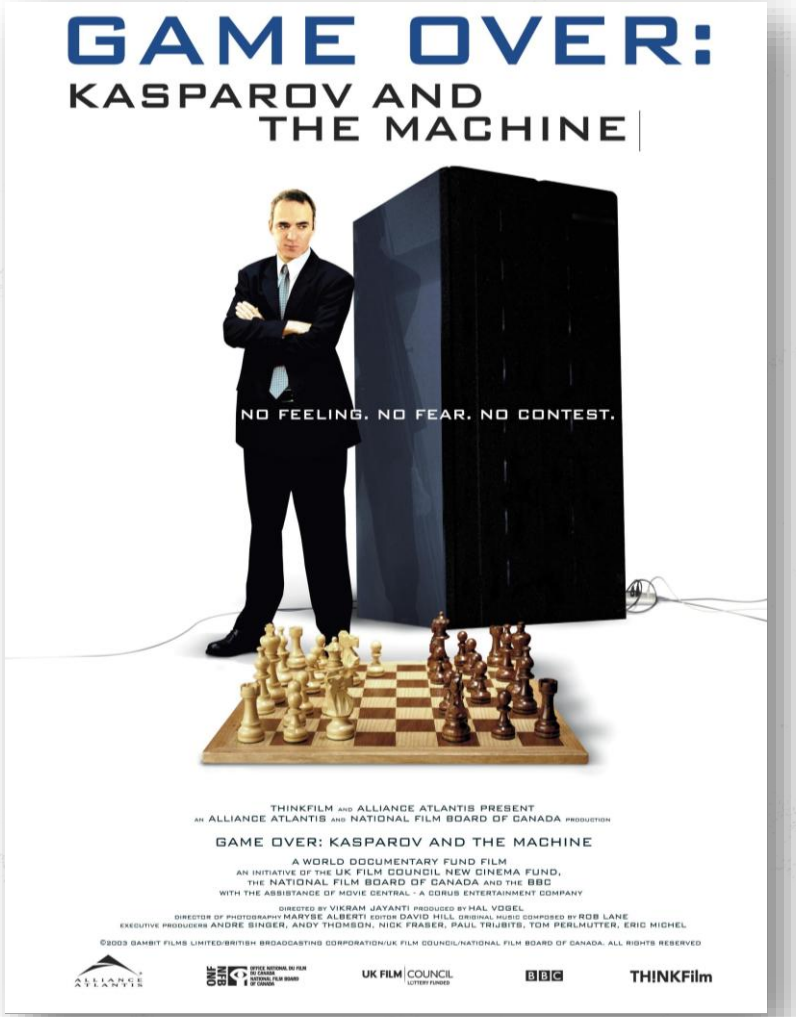


The Second AI Winter (1987–1996)

- The Expert systems couldn’t keep the pace with industry demand and expansion
- The personal computers started flourishing and focus of industry shifted from AI to making compact computers
- The funding agencies started focusing on more pragmatic projects such as the Internet



The IBM Deep-Blue (1997)



https://www.youtube.com/watch?v=NJarxpYyoFI

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The New Millennium











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The New Millennium

- The DARPA Grand Challenge
  - Autonomous Driving



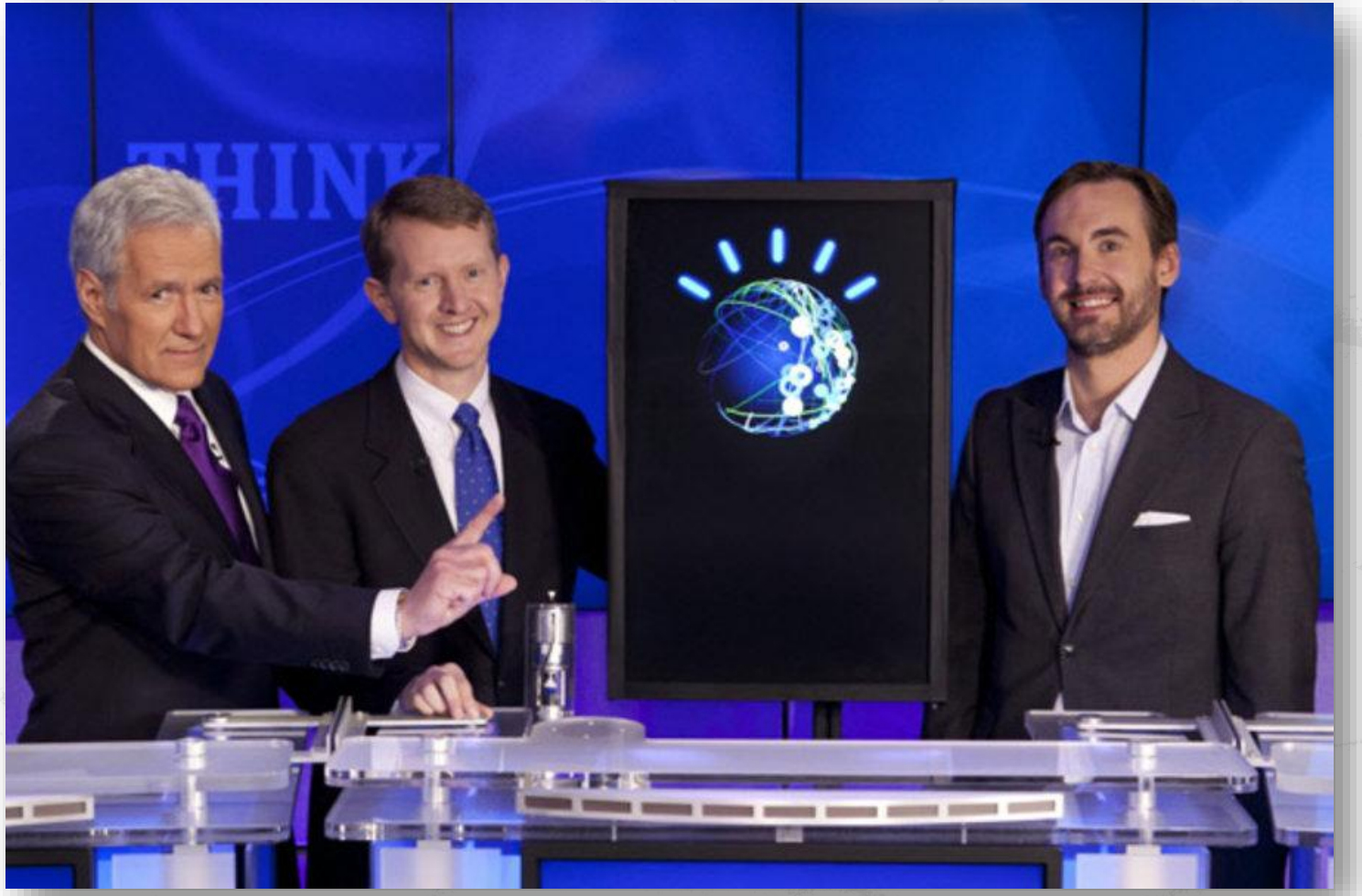
Stanley by Stanford University
Won the DARPA Grand Challenge **2005**



Winner of **2007** Urban Challenge,
Designed by CMU

The New Millennium

IBM Watson wins the Jeopardy game against greatest champions (2011)



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The era of drones



https://www.ted.com/talks/vijay_kumar_robots_that_fly_and_cooperate?language=en#t-926912

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The New Millennium

In 2016, Google's DeepMind Project AlphaGo defeats world's Go Champion Lee Sedol in the board game

19×19 board: ~2.082×10^{170} (i.e., a 2 followed by 170 zeroes)



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The New Millennium

- In October 2017, Sophia became the first robot to receive citizenship of any country.
- In November 2017, Sophia was named the United Nations Development Programme's first ever Innovation Champion, and is the first non-human to be given any United Nations title.



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2019 Turing Award Won by 3 Pioneers in Artificial Intelligence

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From left, Yann LeCun, Geoffrey Hinton and Yoshua Bengio. The researchers worked on key developments for neural networks, which are reshaping how computer systems are built.

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ChatGPT



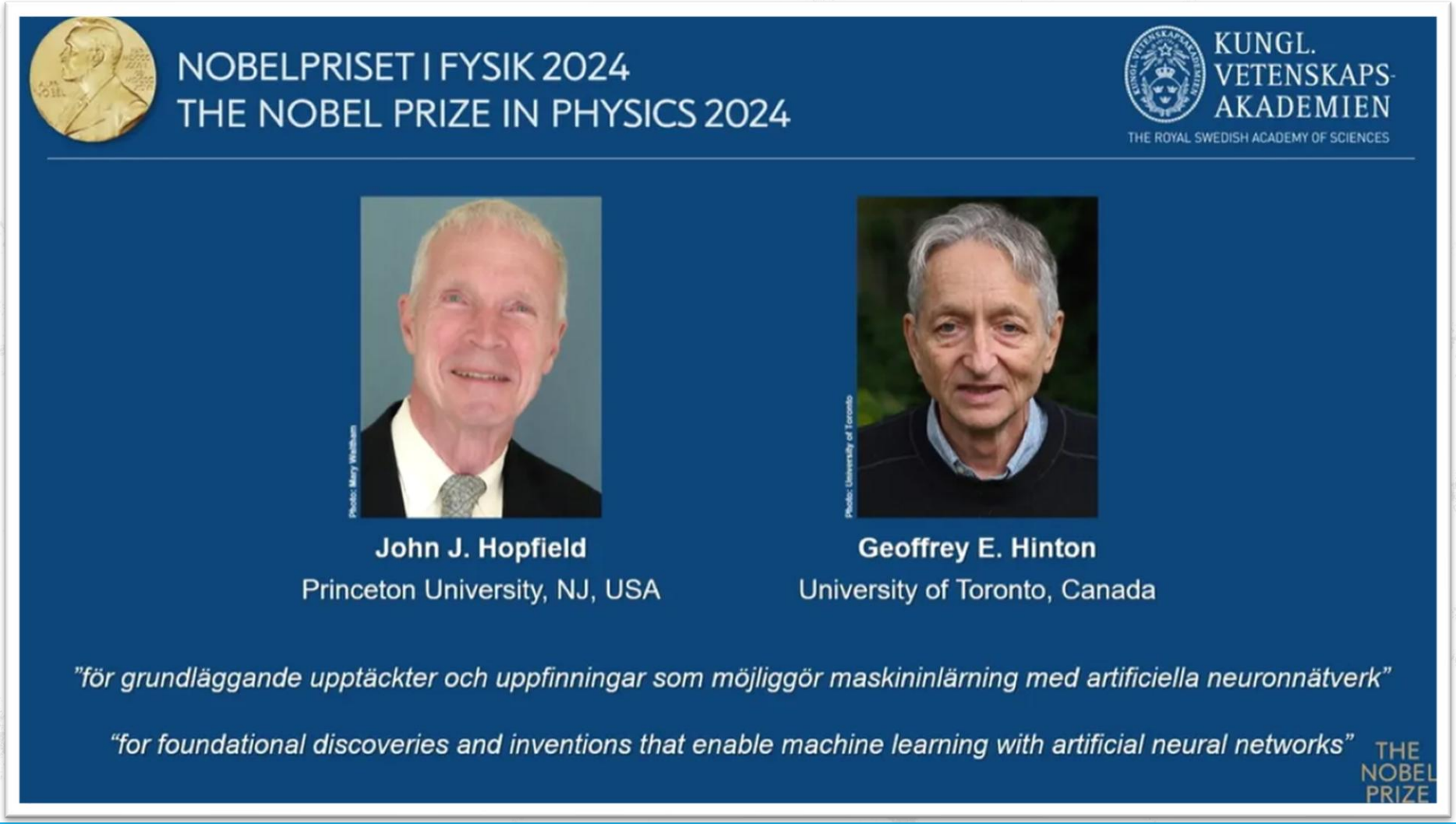
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- **ChatGPT** (Chat Generative Pre-Trained Transformer)*
  - A large language model (LLM)-based chatbot developed by **OpenAI**
  - Launched on **November 30, 2022**,
  - Notable for enabling users to refine and steer a conversation towards a desired length, format, style, level of detail, and language used.
  - Successive prompts and replies, known as **prompt engineering**, are taken into account at each stage of the conversation as a context

*source: Wikipedia

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2024 Nobel Prize in Physics



https://www.nobelprize.org/prizes/physics/2024/summary/

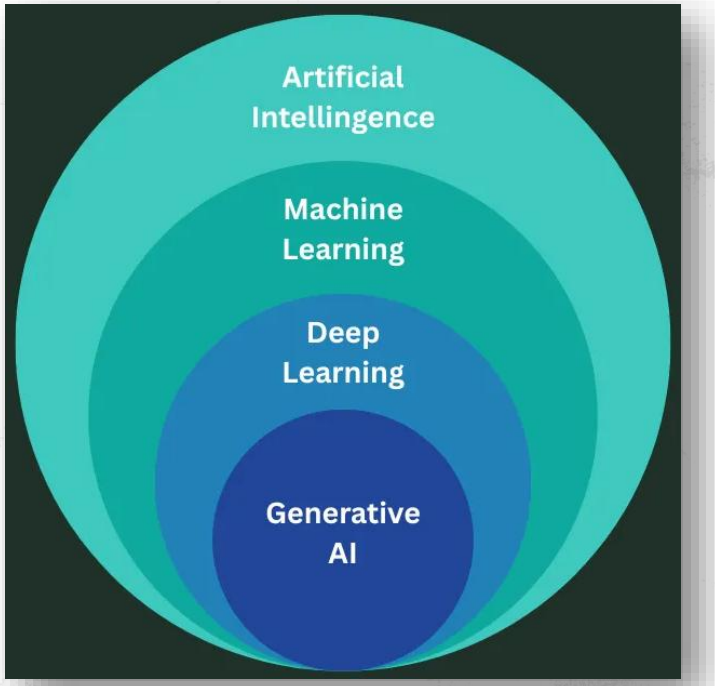
The Broad Scope of AI

- AI spans multiple capability layers:
  - Perception – vision, speech, sensing
  - Prediction – forecasting, risk analysis
  - Reasoning – planning and optimization
  - **Generation** – text, images, code
  - Action – robotics and control systems
- Chatbots represent only a small visible part of this ecosystem.

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Generative AI

- Generative AI (such as ChatGPT, Gemini, etc) focuses on creating content:
  - Text, images, audio, video, code
  - Natural language interaction
- Generative AI is:
  - **A subset of AI**
  - Powerful for communication and creativity
  - Not suitable for all tasks
- It complements—not replaces—other AI systems.



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Risks and Limitations of AI

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- Across all AI systems, key risks include:
  - Bias from historical data
  - Lack of explainability
  - Errors and uncertainty
  - Over-automation
  - Privacy and security concerns
  - Hallucinations



The Near Future of AI

- Likely developments:
  - AI embedded in most digital tools
  - Greater autonomy in systems
  - Increased regulation and standards
  - More private and on-device AI
  - Stronger integration with robotics
- AI will become less visible but more powerful.



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Key Take-Aways

- AI is:
  - Broad
  - Transformative
  - It is going to stay
- But:
  - Humans define goals
  - Humans set values
  - Humans remain responsible

- **AI is a tool** for progress — **not a replacement for human judgment.**

Conclusions

- AI is transforming industries now
- Future AI is multimodal, autonomous, and domain-specific
- Adoption requires responsibility and strategic planning



For Further Learning

- For additional skilling, advanced training in Artificial Intelligence and allied domains, please reach out to NIELIT.
- Visit **www.nielit.gov.in** or contact your nearest NIELIT Centre for guidance and support.

धन्यवाद



THANK YOU